Pharmacology of the Adrenal Gland Part 1

Monzurul A. Roni, Ph.D.

Block 7 Independent Study

Acknowledgment: Linda Chang, Pharm. D.; Stephen Lasley, Ph.D.

MATERIALS INTENDED FOR UNIVERSITY OF ILLINOIS STUDENTS ONLY. DO NOT DISTRIBUTE.

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Welcome to block 7 ind study titled pharmacology of adrenal glands. This is Dr. Roni. In this two part video series, we will cover important drugs for the treatment of hypo or hyperactivity of adrenal hormones. At the end of this video, we will go over some practice questions.

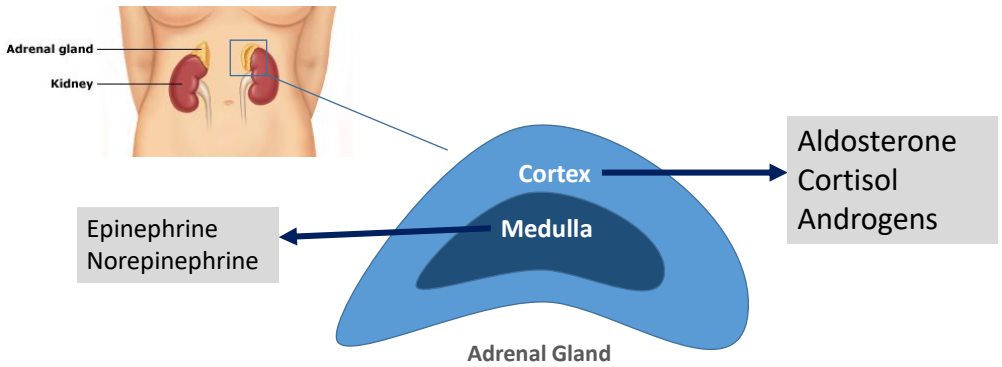
Objectives

1. Discuss the pharmacological management of adrenal insufficiency.
2. Describe the mechanisms of action, therapeutic uses, and adverse side effects of glucocorticoids.
3. Compare and contrast short, intermediate, and long acting glucocorticoids with regard to anti-inflammatory potency, sodium-retaining potency, and biological half-life.
4. Describe the rationale for alternate-day drug therapy for patients on long-term glucocorticoids. Specify the need for gradual withdrawal from chronic glucocorticoid therapy and list the factors determining rate of dose reduction.

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

After watching this video, you will learn about adrenal insufficiency and the role of glucocorticoids as a treatment. Specially, we will talk about synthetic Corticosteroids which are imp class of drugs and high yield for step exams.

Adrenal Glands

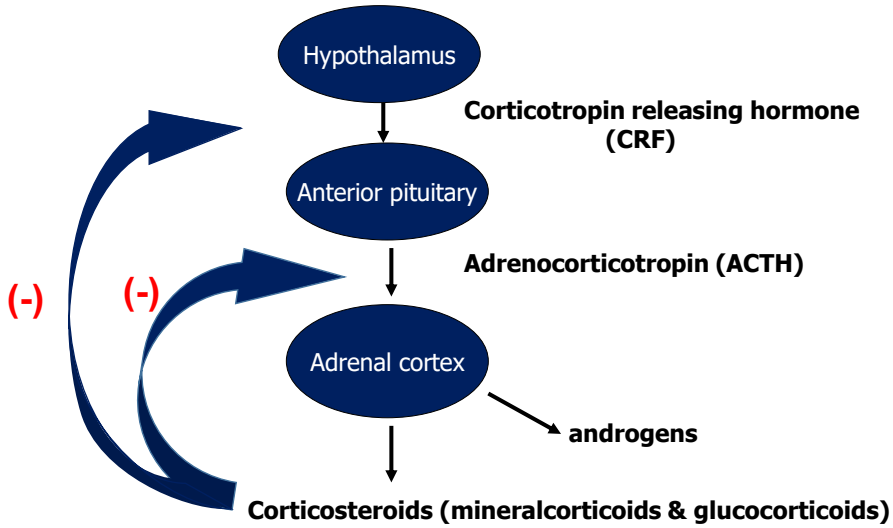


UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

3

Before talking about drugs, I would like to step back and talk little bit about the adrenal hormones. These hormones are produced by adrenal glands, located on top of our kidneys. The outer layers of the gland, known collectively as adrenal cortex, release 3 types of hormones- aldosterone, corticol, and androgens. Aldosterone or mineralocorticoid plays important role in Na ion retention, cortisol or glucocorticoid plays important role in metabolic and immune system, and androgens are primarily important for sexual characteristics.
The medulla, on the other hand, releases epinephrine and to lesser extent norepinephrine hormones.

Regulation of Adrenal Hormones



UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

The secretion of corticosteroids are regulated by the Hypothalamic-Pituitary-Adrenal Axis. Corticotropin releasing hormone secreted by the hypothalamus stimulates anterior pituitary to release adrenocorticotropin hormone which ultimately stimulates adrenal cortex to release corticosteroids. If we have a reduced secretion of ACTH, the production of corticosteroids falls- leading to adrenal insufficiency. On the other hand, increased ACTH and increased corticosteroid secretion triggers the negative feedback mechanism, which will also suppress the release of steroid hormones.

Adrenal Insufficiency

↓ Cortisol and aldosterone production

Weakness, fatigue
Weight loss

Hyponatremia
Hyperkalemia
Orthostatic hypotension
Salt craving

↑ ACTH
1° insufficiency

↓ ACTH
2/3° insufficiency

Muscle ache
GI disturbances
Sugar craving
Metabolic acidosis
Skin/mucosal hyperpigmentation



Source: First Aid for USMLE Step1

Addison disease

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

5

The hyposecretion of glucocorticoid and mineralocorticoid give rise to Adrenal insufficiency. Primary insufficiency results from increased production of ACTH, whereas secondary/tertiary insufficiency is related to decreased ACTH production.

The primary causes of adrenal insufficiency are autoimmune disease, infection, or metastatic cancer. The secondary causes could be due to **Long-term glucocorticoid administration ,** Lack of adrenal stimulators such as ACTH or Tumors.

The common clinical signs and symptoms are weakness, fatigue and weight loss.

Due to lack of mineralocorticoids, our body fails to retain Na ions. Therefore, Na as well as water level decreases in the blood- leading to low blood pressure. Other symptoms include hyperpigmentation of skin and gum.

Chronic adrenal insufficiency is known as Addison disease.

Pharmacological Management

LO1

Cortisol replacement

- Hydrocortisone or cortisone acetate
- Other corticosteroids option
  - Prednisone or dexamethasone
- Dose adjusted based on urinary cortisol level
- Drug interaction
  - Larger dose may be needed with enzyme inducers

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Cortisol replacement

Hydrocortisone or cortisone acetate

Option 1

15mg in the morning, 10mg in the afternoon

Option 2

TID dosing (awakening, breakfast, evening meal)

Other corticosteroids option

Prednisone 5mg/d or dexamethasone 0.5mg/d

Enzyme inducers: barbiturates, carbamazepine, rifampin etc.

Pharmacological Management

LO1

Mineralocorticoid replacement

- Fludrocortisone acetate
  - Dose adjusted based on potassium level and blood pressure

- Extra supplements
  - Periods of acute illness, stress, or surgery
  - ⇑ hydrocortisone

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

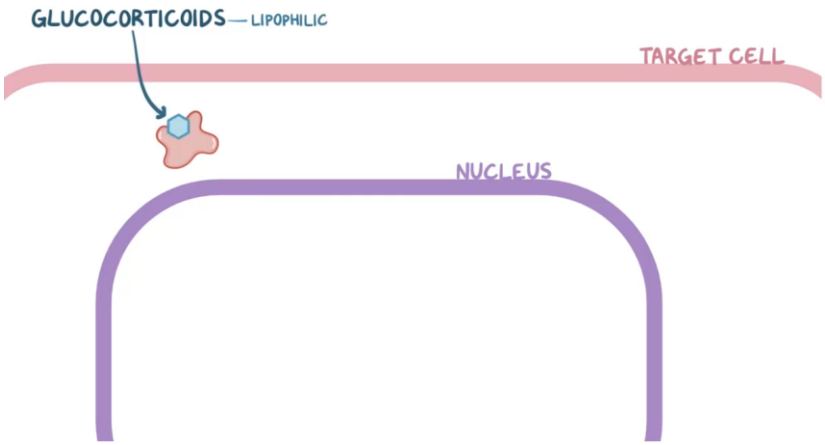
Fludrocortisone acetate

A potent synthetic mineralcorticoid
50-200mcg/d
Adjust dose bases on potassium level and blood pressure

Additional glucocorticoids can be given in Periods of acute illness, stress, or surgery
⇑ hydrocortisone to 75-100mg/day (normal level is increased in these conditions)

MOA of Glucocorticoids

LO2



UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

"Glucocorticoids" by Osmosis

Now lets focus on how glucocorticoids act in the body.
Glucocorticoid receptors are located inside the cell. Due to their lipophilic nature, steroid drugs easily cross the lipid bilayer of cell membranes and bind to intracellular receptors. Upon binding, the receptor dimerizes and actively transported into the nucleus, where it interacts with DNA and nuclear protein. It binds to glucocorticoid receptor elements or GRE in the promoters of responsive genes which regulates transcription of mRNA for the production of proteins that brings about the final response.
Alternatively, steroid receptor complex can interact with and alter the function of other transcription factors such as Nf-kB in the nucleus. These transcription factors mediate anti-inflammatory and immunosuppressive effects of glucocorticoids.

Therapeutic Indications of Glucocorticoids

LO2

- Replacement therapy
- Anti-inflammatory activity
- Immunosuppressive effects

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

➢Apart from hormone replacement therapy, steroids are very useful for treating inflammatory conditions such as lupus, RA, IBD, UC, conjunctivitis, dermatitis, rhinitis, and asthma to name a few.
➢They are also used to suppress immune response during organ transplantation.

Selection of Agent

LO3

Glucocorticoid	Equivalent Dose (mg)	Anti-inflammatory potency	Mineralocorticoid potency	Duration of action
Cortisone	25	0.8	2	S
Hydrocortisone	20	1	2	S
Prednisone	5	4	1	I
Triamicinolone	4	5	0	I
Methylprednisone	4	5	0	I
Dexamethasone	0.75	20-30	0	L
Betamethasone	0.6	20-30	0	L

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

If we compare the glucocorticoids available on the market, we will find they differ on potency and duration of action.
S = short acting (8-12 hours biological half-life)
I = intermediate (12-36 hours biological half-life)
L = long (36-72 hours biological half-life)

Biological half-life is much longer than their plasma half-life

Adverse Effects

LO2

-**The longer the corticosteroid use, the greater the risk for side effects**

-IV/IM > Oral >> inhaled formulations>>>topicals

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Unfortunately, steroids have long list of possible adverse effects. adverse effects can appear from Continued use of supraphysiological doses or from sudden cessation of therapy.

Adverse Effects

LO2

- Aldosterone like effect
  - ⇑ Na and ⇓ K
- Fluid retention and hypertension

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Side effects like sodium retention,
Hypokalemia,
Fluid retention and hypertension are
Commonly seen in patients with primary hyperaldosteronism and in patients treated with potent mineralocorticoids,

Adverse Effects

LO2

- Immune response
  - Inhibits the immune system and the inflammatory response,
  - ⇑ susceptibility to infection
- Hematological
  - Lymphocytopenia
  - Eosinopenia,
  - Monocytopenia
  - Neutrophilia

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Since Glucocorticoids suppress inflammatory cytokines and chemokines and other mediators of inflammation, the unwanted consequence is increased susceptibility to infection. For e.g. patients taking steroid inhalers may develop oral candidiasis.
Glucocorticoids also reduce circulating lymphocytes, eosinophils, monocytes but increase neutrophils. The increase in neutrophils is due to decreased migration from blood vessels and increased production from bone marrows.

Adverse Effects

LO2

- Peptic ulcer disease
- Myopathy
- Behavioral changes
  - Insomnia, nervousness, changes in mood or overt psychosis
- Cataracts

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Peptic ulcer disease

Possible effects: thinning of gastric mucosa and increased acid production

Myopathy

Weakness of the proximal limb muscles

Cataracts in eye have

Well-established connection to therapy

Adverse Effects

LO2

Osteoporosis

- Inhibits osteoblast activity
- Increases risk for female, RA, DM, immobilized patient

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Preventive therapy:

Calcium supplement (1500mg/day)
Vitamin D
Weight-bearing exercise
Biphosphonates: alendronate, risedronate

Adverse Effects

LO2

- Hyperlipidemia
  - Increases total cholesterol, LDL, and TG
- Hyperglycemia

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Hyperglycemia

Stimulates gluconeogenesis and inhibits peripheral utilization of glucose

Promotes glycogenolysis by stimulating glucagon release

Increased glucose level seen within 2 weeks

Prevention

Educate patient on symptoms of DM

Prevention of Adverse Effects

LO2

- A diet low in sodium chloride and high in potassium
- A diet restricted in calories to prevent weight gain
- Antacids – three or four times daily
- Supplementary vitamin D
- Physiotherapy – increased active muscular exercise as tolerated

Adverse Effect Management

LO4

- Lowest effective dose
- Do not stop chronic therapy abruptly

- Alternative day therapy (QOD)
  - Recommended for prolonged therapy
  - Uses large dose of intermediate acting agent in the morning

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

To reduce the adverse effects we have to use-

Alternative day therapy (QOD)

Recommended for prolonged therapy

Uses large dose of intermediate acting agent in the morning

Advantages

Decreased HPA suppression

Decreased Cushingoid side effects

Question

- A 49 y/o woman with rheumatoid arthritis involving joints of both hands and hips has developed an exacerbation of her disease in the past few weeks. She has been receiving methotrexate for the past 6 months. Her provider is starting her on prednisone 5 mg therapy.

- Why was prednisone chosen for this patient?
- What are the potential side effect we can anticipate in this patient in the next 2 weeks?

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

19

The table shows the potencies and anti-inflammatory and mineralcorticoid effects for various agents. Prednisone retains a good balance between anti-inflammatory effect and min. mineralocorticoid side effects.

Fluid retention and hypertension

Hyperglycemia

nervousness and has the difficulty sleeping

Question

- After 3 months of prednisone therapy, patient does not like the weight gain and self discontinue prednisone therapy.
- What potential adverse effects do we anticipate for this patient?

Our biggest fear is acute adrenal insufficiency symptoms (nausea, vomiting, abd pain, hyponatremia, hyperkalemia, weakness, lethargy, hypotension).

Question

A 46 y/o woman with Crohn's disease has been taking dexamethasone daily for the past 1 year. She has been experiencing a thickness in the back of her neck. Which of the following best explains this symptom?

A. Edema
B. Tumor
C. Increase in muscle growth
D. Redistribution of lipid

UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

21

answer: D

Dexamethasone lacks mineralocort activity so water retention and edema is not the answer.

Question

A 15 y/o male with chronic pulmonary disease is being treated with inhaled beclomethasone. He is likely to experience which of the following side effects?

A. Buffalo hump
B. Acute adrenal insufficiency
C. Hyperglycemia
D. Oral candidiasis

Answer D

Topical beclomethasone reduce the immune response, so yeast infection/oral thrush is common. The patient should rinse mouth and spit after inhaler use. Systemic s/e are not common for topical steroids.

This is the end of part 1. In part 2, we will discuss about abnormally high level of corticosteroids and its treatment.